



Intelligent Chinese patent medicine (CPM) recommendation framework: Integrating large language models, retrieval-augmented generation, and the largest CPM dataset

Suyang Qin^{a,1}, Yifan Wang^{b,1}, Tangming Cui^{a,1}, Jing Ma^a, Xin Zhou^c, Xinglin Guo^d, Chuchu Zhang^a, Chongyun Zhou^a, Rongjuan Guo^{e,*}, Haiyan Li^{a,**}

^a Institute of Information on Traditional Chinese Medicine, China Academy of Chinese Medical Sciences, Beijing 100700, China

^b School of Basic Medical Sciences Peking University, Beijing 100191, China

^c Huzhou Traditional Chinese Medicine Hospital Affiliated to Zhejiang Chinese Medical University, Huzhou, Zhejiang 313000, China

^d First Teaching Hospital of Tianjin University of Traditional Chinese Medicine, National Clinical Research Center for Chinese Medicine Acupuncture and Moxibustion, Tianjin 300381, China

^e Psychosomatic Department, Dongfang Hospital, Beijing University of Chinese Medicine, Beijing 100078, China

ARTICLE INFO

Keywords:

Large language models (LLMs)
Multi-LLMs Validation
Chinese patent medicine (CPM) dataset
retrieval-augmented generation
intelligent CPM recommendation framework
guidelines evaluation

ABSTRACT

Chinese patent medicines (CPMs), a vital component of healthcare systems in China and worldwide, has been increasingly utilized in clinical practice. However, approximately 70 % of CPMs are prescribed by Western medicine physicians who lack expertise in traditional Chinese medicine syndrome differentiation and treatment. Here, in this study, we constructed RAG-CPMF, an intelligent CPM recommendation framework integrating large language models (LLMs), retrieval-augmented generation (RAG), and the largest CPM dataset. Specifically, we first applied the Multi-LLMs Validation method, significantly reducing the manual proofreading workload involved in structured data extraction. Furthermore, based on this approach, we successfully established the largest CPM dataset (<https://gitee.com/tcmdoc/cpm>) with continuous on-line updates. Afterwards, we combined the CPM dataset and the RAG architecture, along with effective integration of generative capacity from LLMs and external data retrieval; all of which contributed to the RAG-CPMF. Finally, the accuracy of RAG-CPMF was evaluated against clinical guidelines, demonstrating that this framework significantly improved CPM recommendation accuracy compared with general-purpose LLMs.

1. Introduction

Traditional Chinese medicine (TCM), in the long term, has demonstrated outstanding effectiveness in improving health and preventing diseases [1,2]. Moreover, TCM theory can offer valuable resources and inspiration to the development of modern medicine. For example, the extraction of artemisinin (Qing Hao Su) from *Artemisia annua* has contributed to the groundbreaking progress in malaria treatment, saving

millions of lives [3,4]. Based on it, under the guideline of TCM theory, Chinese patent medicines (CPMs) refer to TCM preparations with the combination of Chinese herbs and modern pharmaceutical technology, commonly in convenient forms like capsules, granules [2]. Compared with TCM decoctions, CPMs are more convenient in administration and storage, as well as play a crucial role in the treatment, prevention, and rehabilitation of diseases. Today, CPMs have been an essential component of pharmaceutical healthcare systems in China, even in the world,

Abbreviations: AI, Artificial Intelligence; API, Application Programming Interface; CoT, chain-of-thought; CPMs, Chinese patent medicines; FoT, forest-of-thought; LLMs, large language models; NMPA, National Medical Products Administration; RAG, retrieval-augmented generation; TCM, traditional Chinese medicine; ToT, tree-of-thought.

* Correspondence to: Psychosomatic Department, Dongfang Hospital, Beijing University of Chinese Medicine, No.6 Fangxingyuan Community, Fengtai District, Beijing 100078, China.

** Correspondence to: Institute of Information on Traditional Chinese Medicine, China Academy of Chinese Medical Sciences, No.16 Nanxiaojie, Dongzhimen nei, Dongcheng District, Beijing 100700, China.

E-mail addresses: dfguorongjuan@163.com (R. Guo), li_haiyan01@126.com (H. Li).

¹ These authors contributed equally: Suyang Qin, Yifan Wang and Tangming Cui

<https://doi.org/10.1016/j.phrs.2025.107883>

Available online 24 July 2025

1043-6618/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

experiencing increasingly widespread application in clinical practice [5]. For instance, Tongxinluo capsules have been shown to improve clinical outcomes in patients with ST-segment elevation myocardial infarction [6].

It is indicated by survey that approximately 70 % of CPMs in China are prescribed by western medical physicians and are used in combination with chemical drugs to enhance efficacy and reduce side effects [7]. However, in clinical practice, these prescriptions are frequently based solely on disease categories, while ignore the fact that a certain disease may correspond to multiple CPM treatment options [8]. This kind of pattern would not only result in potential risks but also reduce the efficacy of CPM treatments [7]. In contrast, TCM, whether in diagnosis or in treatment, will be based on patients' specific symptom, then prescribing corresponding CPMs [9,10]. Recent studies have shown that symptom-related gene modules exhibit significant proximity to the targets of active herbal compounds, offering systems biology support for symptom-based drug selection in TCM [11]. In view of this, it is necessary to develop an intelligent framework that recommends CPMs based on patient symptoms, providing decision support aligned with TCM theory for Western medicine physicians.

Large language models (LLMs) are generative artificial intelligence (AI) models based on transformer architecture, with billions of parameters [12]. In healthcare, LLMs can interpret complex medical concepts and provide substantial support for medical examinations, assisted diagnosis, treatment prediction, and clinical report generation [13–17]. However, since LLMs are generally trained on publicly available data without specialized optimization for specialized clinical knowledge, leading to potential “hallucination” risks in clinical recommendations [18–21]. For example, the clinical advice generated by LLMs is typically only about 60 percent consistent with established clinical guidelines [22,23].

Retrieval-augmented generation (RAG) is an architecture designed to enhance the capabilities of LLMs by incorporating an external information retrieval [24]. Its advantage lies in leveraging external knowledge bases to provide foundational information to LLMs, thereby significantly reducing “hallucinations”, improving accuracy, and effectively avoiding the high costs associated with extensive fine-tuning [25]. As an external knowledge base within the RAG framework, the quality of the CPM dataset critically influences the performance of the CPM recommendation framework. Currently, CPM-related knowledge is scattered and lacks structured format, posing challenges for RAG-based

external information retrieval. Consequently, constructing a high-quality, comprehensive CPM dataset covering all medicines is an urgent necessity.

This study proposed a CPM recommendation framework (RAG-CPMF) based on symptom information, which supported multiple input formats. The framework first converted inputs into standardized TCM clinical terms, then performed similarity matching within the CPM dataset to retrieve relevant instruction information. This information was subsequently incorporated into an LLM to generate the final CPM recommendations (Fig. 1). To support the framework, with the foundation of CPM instructions, this study employed Multi-LLMs Validation and constructed a large-scale dataset of 8644 CPMs with key information such as prescription, indications, and side effects. Based on this, the study introduced a RAG architecture and used the CPM dataset as an external knowledge base. It retrieved CPM-related information highly relevant to the input symptoms and incorporated it into the context for the LLM, significantly improving the precision and reliability of the recommendations. In addition, by integrating a TCM clinical terminology system, the framework can automatically convert natural language expressions and TCM synonyms into standardized TCM terms [26]. (See <http://tcmcts.cintcm.com/login> for details.) We employed Qwen2.5-72B-Instruct, the largest parameter version of the top-performing model on the TCM exam as of October 2024 [27].

2. Methods

2.1. Data collection and processing

This study utilized the national Drug Code Information dataset from the National Medical Products Administration (NMPA) of China [<https://www.nmpa.gov.cn/>], which provides the up to date and the most complete information of domestic drugs. This dataset covers various types of drug records, including Chinese medicines (labeled as Z), ancient classical compound preparations (labeled as C), chemical drugs (labeled as H), biological products (labeled as S), and health products (labeled as B). The columns of the dataset consist of the national approval number, pharmaceutical name, dosage form, specification, marketing authorization holder, manufacturer, and national drug code.

To comprehensively collect the pharmaceutical names of CPMs, we preprocessed the dataset as follows. First, we consolidated drugs with the same pharmaceutical name but from different manufacturers. Next,

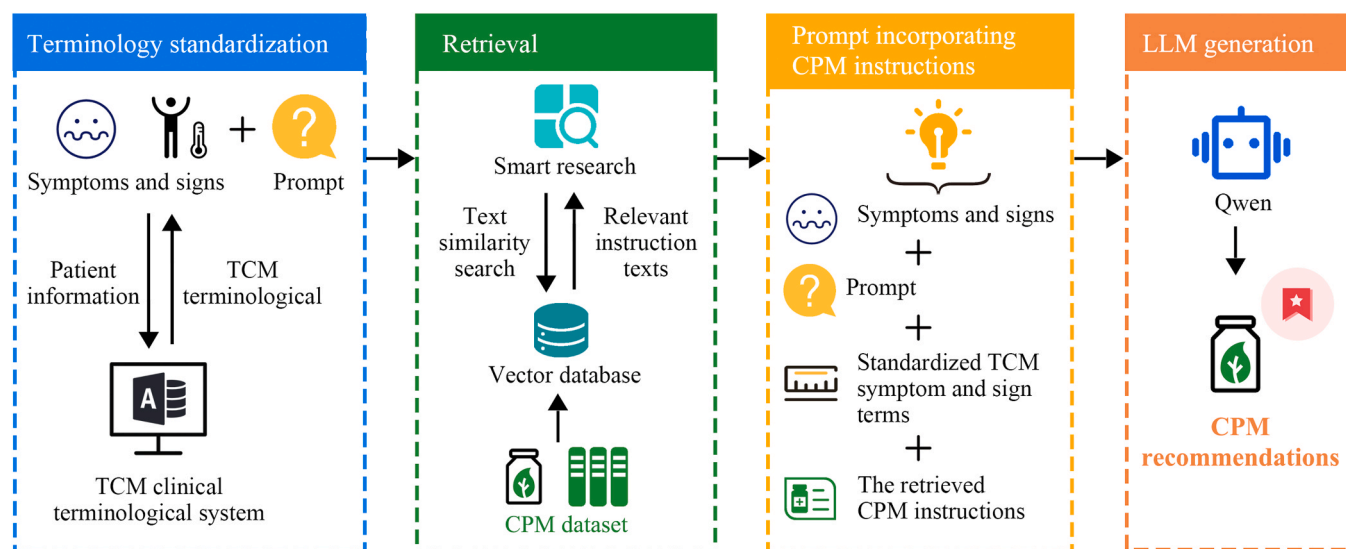


Fig. 1. Intelligent Chinese patent medicine recommendations based on retrieval-augmented generation technology. The framework converts symptom inputs into standardized TCM clinical terms, retrieves relevant information from a CPM dataset using similarity matching, and incorporates this information into a LLM to generate final CPM recommendations.

we removed drug records labeled as ‘B’, ‘H’, ‘S’, and ‘X’, leaving Chinese medicines and ancient classical compound preparations labeled as ‘Z’ and ‘C’. Furthermore, active pharmaceutical ingredients were excluded, as they serve only as components in the production of CPMs, not the final drugs. After these processing steps, we obtained a dataset of unique CPM records.

In order to ensure the completeness and accuracy of the data, we collected the CPM instructions and used the contents to supplement the dataset. We retrieved data from multiple official websites, including those of various manufacturers, online pharmacies, the NMPA, and the National Administration of Traditional Chinese Medicine. From these sources, we have collected the latest versions of Chinese patent medicine instructions available in either text or image format.

2.2. Data records

The dataset is designed to assist healthcare professionals in quickly retrieving and comparing different drugs, meanwhile providing personalized drug recommendations based on the specific symptoms of patients. To enhance its comprehensiveness, we have added the following columns based on official documents from the NMPA and evidence-based clinical practice guidelines for CPMs: 1) prescription of CPM, 2) indications or functions, 3) indications or functions of Ethnic Patent Medicine, 4) dosage and usage, 5) side effects, 6) contraindications, 7) description, 8) precautions and warnings, 9) pharmacologic action. The full column names and descriptions of the dataset are shown in Table 1.

2.3. Construction of CPM dataset based on Multi-LLMs Validation

Despite the fact that the new fields can be found within instructions texts, their unstructured nature prevents direct access to the dataset. Manual data extraction would be a labor-intensive task, which needs an automatic data processing framework to extract important information from the texts and export it in a structured text. Systematic review guidelines note that single person data extraction can lead to omissions and errors, affecting data quality. To address this, dual data extraction with independent cross-validation by multiple investigators is recommended to enhance accuracy and reduce bias [28]. Inspired by this, we have come up with a Multi-LLMs Validation method which would take advantage of multiple LLMs operating independently to extract structured information, followed by cross-model consistency validation to ensure the accuracy of results and significantly reduce the workload of manual proofreading.

The extraction process from the instructions texts is divided into three stages, namely sampling and testing, hypothesis proposal and application, and technical validation (Fig. 2). In the first place, we conduct a sampling test to manually evaluate the accuracy of the Multi-LLMs Validation method. Then we would raise a certain hypothesis, that is, provided at least n LLMs can offer consistent outputs which means the accuracy reaches 100 %, we will have no need of manual proofreading. Based on such hypothesis, it is concluded that when at least n LLMs provide consistent answers, the number of texts involved is x . In other words, it is assumed that the accuracy of the structured information extracted from x texts is 100 % without manual proofreading. Finally, to evaluate their accuracy and validate our hypothesis, we perform a random sampling and manual proofreading of x texts.

2.3.1. Model setting

In this study, we selected five LLMs: Qwen [29], GLM [30], ERNIE [31], Llama [32], and ChatGPT [33], which have been the largest parameter versions of the latest models released by their respective developers as of October 2024, with details shown in Table 2. All models were accessed via the Application Programming Interface (API) using Python (version 3.12), and the temperature parameter was consistently set to 0.01.

Table 1
Description of each column name in the Chinese patent medicine (CPM) dataset.

Column Name	Description
Approval Number	The drug approval number serves as a distinctive identifier assigned to a drug that has been reviewed and approved by National Medical Products Administration (NMPA), in accordance with Law of the People's Republic of China on the Administration of Pharmaceuticals. With uniqueness, numbers can confirm that aspects, including manufacturing technologies, ingredients, instructions, finally meet a criterion of NMPA. Based on the above, approval numbers play a vital role in production, access to market, as well as circulation.
Pharmaceutical Name	It is the official name assigned to a drug or medication by NMPA.
Dosage Form	It is regarded as the physical form of a prepared medicine which has been tested by clinical use and manufactured in accordance with standard technical procedures, designed for medical treatment or prevention. In light of different routes of medication administration, there are various kinds of standardized Chinese medical preparations derived from traditional Chinese herbal formulas, such as granules, powder, medicated paste, etc.
Specification	Specification, preparation unit, is determined by usage and dosage of drugs. It is often expressed as content of effective elements in drugs per unit.
Marketing Authorization Holder	A drug marketing authorisation holder is an institution or legal entity licensed to product, distribute and commercialize their pharmaceutical products on the premise that they have granted a drug registration certificate and marketing authorisation. A marketing authorization holder must take responsibility for drug safety, effectiveness and quality controllability under the guidance of Law of the People's Republic of China on the Administration of Pharmaceuticals in China.
Manufacturer	Enterprises or organizations with the approval acquired from drug regulatory departments of their corresponding provinces, autonomous regions, and municipalities shall obtain the drug production license beforehand. It refers to the entities who are authorized to engage in drug production activities via the above process.
National Drug Code	It refers to the code comprising figures, based on pharmaceutical information, which is issued by NMPA in China. The unique code is related to manufacturer, approval number or registration certificate number, as well as variety, dosage form and specification.
Prescription of CPM	The prescriptions of CPMs consist of relevant ingredients and traditional Chinese medicine or Chinese herb extracts. It is a kind of therapy with guidelines of Traditional Chinese medicine theory, following compatibility of medicines.
Indications or Functions	Functions are typically described as treatment methods and therapies. Indications usually refer to the specific situations or symptoms that drugs are available to prevent, treat and cure.
Indications or Functions of Ethnic Patent Medicine	Ethnomedicine, also named as traditional medicine, originates from and is used in various minorities in China. It is the drug that minorities apply in line with the practical experience or theories of traditional medicine from their groups. They have developed distinctive terminologies of symptoms and situations, as well as corresponding treatment methods and therapies, tailoring to their own cultural contexts.
Dose and Usage	The route, dose and frequency of medication administration.
Side Effects	Side effects encompass any unintended clinic effects of drugs within the therapeutic range,

(continued on next page)

Table 1 (continued)

Column Name	Description
Contraindications	whether or not these effects are adverse or harmful. There are two types of contraindications in instructions: ban and caution. The former, the strongest warning, means that medicine is absolutely forbidden. If not, it will lead to serious results like life-threatening consequences; while the latter tells that drugs should be used with caution in order to prevent side effects.
Description	Description indicates the physical and chemical characteristics or appearance of the drug, which are typically detailed in terms of the color, odor, form, solubility, and other relevant attributes.
Precautions and Warnings	Certain considerations, while taking medicine, are to ensure the safety of drug use.
Pharmacologic Action	It refers to how medication interacts with the human body and its effects. Among the processes are absorption, degradation, metabolism and excretion of drugs.

2.3.2. Prompt design

Prompt engineering improves the performance of LLMs in structured information extraction tasks. Meanwhile, prompt design plays a key role in effectiveness [34,35]. At present, prompting strategies often include Chain of Thought (CoT) and Tree of Thought (ToT). CoT prompts guide LLMs, through a linear reasoning path, to break down complex tasks into ordered steps [36]. ToT prompts adopt a tree-like structure to explore multiple reasoning paths and select the optimal solution through a pruning mechanism [37].

Further research on prompting strategies has shown that reasonably decomposing complex tasks into a series of subtasks can significantly improve the accuracy of LLMs in complex reasoning tasks [38]. Inspired by these approaches, we further proposed Forest of Thought (FoT) prompting. FoT breaks the overall task into several subtasks, each initialized with an independent starting point, and each subtask is handled by its own ToT for information extraction. Specifically, CoT and ToT use a single prompt to extract all fields, while each tree in FoT focuses on extracting one specific field and then combines the outputs from all the trees to generate the final answer. An overview of the above can be found in Fig. 3 and Table 3, along with brief descriptions. To evaluate the accuracy of the three prompts, we randomly selected samples from all the instruction texts based on sample size estimation. Afterwards, we have performed structured information extraction from these samples on each LLM using these prompts.

2.3.3. Sampling and testing

After excluding the samples extracted in the previous trial, we randomly selected from the remaining texts to evaluate the accuracy of the Multi-LLMs Validation method in extracting structured information. In this test, we took advantage of independent random sampling without replacement to avoid potential bias resulting from data overlap. The specific steps of the Multi-LLMs Validation method were as follows. First, five LLMs independently extracted structured information from the texts. Next, the extraction results from each LLM underwent cross-model consistency validation, and the number of votes for each result was counted. For different statistical outcomes, the outputs were matched to three predefined rules respectively. The corresponding rules for each outcome were as follows (Supplementary Material 1):

(1) When one result received more votes than any other, we adopted the majority voting strategy from ensemble methods and selected this result as the final output. Previous studies have shown that majority voting strategies can improve overall accuracy [39].

(2) Provided there were three different outputs, two of which were supported by two LLMs respectively and the third by only one, in this case, we would calculate the cosine similarity between the two consistent outputs and the isolated one; afterwards, the group with higher similarity to the isolated result was selected as the final output. Concerning cosine similarity, it can effectively measure the semantic similarity between texts; a higher cosine similarity indicates a smaller semantic deviation [40]. This strategy was a heuristic extension of the majority voting, where cosine similarity was used to treat the isolated result as an “implicit vote”. When one group of two votes was semantically closer to the isolated result, it could be regarded as a scenario with “two explicit votes plus one implicit vote”, effectively forming a majority of three. It should be noted that this heuristic strategy was applied in cases where the voting results were tied. Currently, there is no evidence that a majority result with higher semantic similarity necessarily has greater accuracy. This strategy had potential risks: if the isolated result was a significant factual error (a “hallucination”), this method could amplify the error. Therefore, in such cases, manual validation was ultimately required to ensure the accuracy of the results.

Table 2
Details of included large language models.

Model name	Developer	Details	Release Date
Qwen	Alibaba	Qwen2.5-72B-Instruct	September 19, 2024
GLM	Zhipu AI	GLM-4-Plus	August 29, 2024
ERNIE	Baidu	ERNIE 4.0	October 17, 2023
Llama	Meta	Llama 3.2 90B	September 25, 2024
ChatGPT	OpenAI	ChatGPT-4o	May 13, 2024

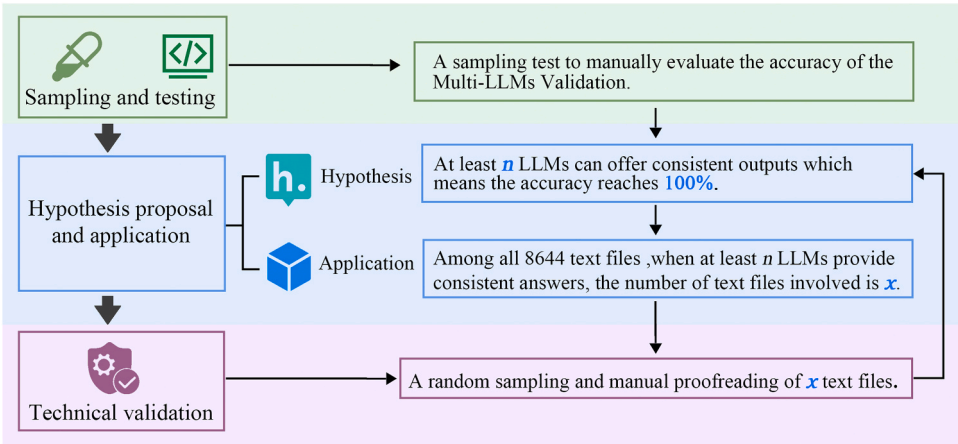


Fig. 2. Three stages of LLMs-based structured information extraction from text. This figure outlines the workflow of structured information extraction using large language models, comprising three stages: sampling and testing, hypothesis proposal and application, and technical validation.

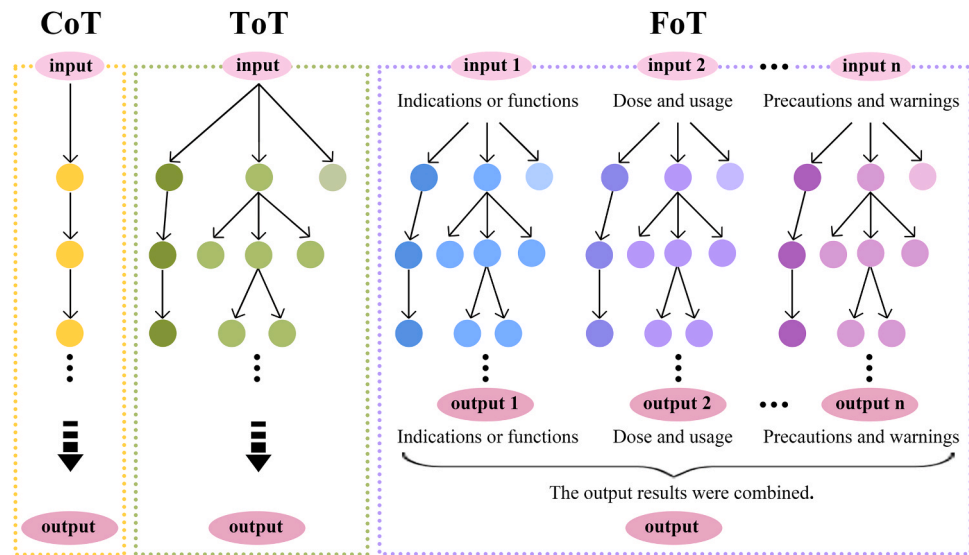


Fig. 3. The schematic diagram of three prompts guiding LLMs to output answers. CoT: chain-of-thought prompting; ToT: tree-of-thought prompting; FoT forest-of-thought prompting. This schematic diagram illustrates the three prompts and the process for evaluating their accuracy in structured information extraction tasks across five LLMs.

Table 3
Descriptions and comparisons of prompting strategies.

Prompting type	Chain-of-thought (CoT)	Tree-of-thought (ToT)	Forest-of-thought (FoT)
Description	Sequential single path reasoning that outputs the result step by step.	A multi-branch reasoning structure that selects the optimal path among several candidates.	The task is divided into multiple subtasks, each processed by an independent reasoning tree, and the results are then aggregated.
Brief explanation	Insert “Think step by step” or a reasoning example into the prompt to guide the model through a predefined reasoning path.	Build a tree-structured reasoning process with multiple branches at decision points, evaluate and select the best path.	Decompose the task into n subtasks, apply a ToT structure to each subtask, and aggregate their results into a final result.
Number of starting points	One	One	Multiple
Reasoning path	Single reasoning path	Multiple reasoning branches from a single starting point	Independent reasoning paths from multiple starting points
Structure type	Single chain (step-by-step)	Single tree (one root, multiple branches)	Multiple trees (multiple roots and branches)

(3) In the event of inconsistent outputs from the five LLMs, all results were retained for manual proofreading.

We evaluated the accuracy of the Multi-LLMs Validation method for structured information extraction by comparing it with manually extracted results. Concretely speaking, two investigators (SQ and TC) manually extracted the structured information from these randomly selected instructions texts, during which any discrepancy was arbitrated by the senior reviewer (HL). Then, we regarded these manually extracted results as the gold standard. The accuracy of the extraction results was determined based on the following criteria: extraction was considered correct if the information extracted from each specification was entirely consistent; any discrepancies were deemed incorrect. Compared with results from the Multi-LLMs Validation method, we could acquire a certain accuracy belonging to consistent results from various combinations of LLMs. Specifically, we assessed the respective accuracy when the results were consistent among five, four, three, and two LLMs.

2.3.4. Hypothesis proposal and application

Based on the results of sampling tests, we proposed a hypothesis, that is, provided at least n LLMs can offer consistent outputs which means the accuracy reaches 100 %, we will have no need of manual proofreading. Subsequently, we adopted the Multi-LLMs Validation method to extract structured information from all instruction texts. Based on this hypothesis, it is concluded that, among all instruction texts, the number of texts involved is x when at least n LLMs provide consistent answers. In

other words, it is assumed that the accuracy of the structured information extracted from x texts is 100 %.

2.3.5. Technical validation

To assess the quality of the dataset extracted by Multi-LLMs Validation method, we have carried out random sampling to instruction texts out of x hypotheses with accuracy of structured information reaching 100 %. In the process, two researchers independently manually extracted structured information from the randomly selected texts and compared the extracted results with those obtained by the Multi-LLMs Validation method.

2.3.6. Comparison of Multi-LLMs Validation with single LLM

We randomly selected samples from all the instruction texts to further verify the practicality and effectiveness of the Multi-LLMs Validation method, comparing it with the traditional approach of extracting structured information using a single LLM. Aiming to reduce experimental variables, the Multi-LLMs Validation method selects a single LLM in common with control groups as the result when the outputs of the five LLMs are inconsistent. For example, when compared with the use of ChatGPT-4o alone, if the results from the five LLMs are inconsistent, the result from ChatGPT-4o is chosen as the final output.

2.4. Intelligent CPM recommendations

We employed the RAG technique to implement intelligent CPM

recommendations. RAG combines the strengths of information retrieval and generative models to effectively improve the relevance and accuracy of generated content [24]. Specifically, RAG first retrieves relevant information from external databases or literature and then integrates this information with a LLM to generate personalized content that aligns with user needs.

In this study, we combined RAG technology with the CPM dataset to develop RAG-CPMF for accurate CPM recommendations. First, in Sections 2.1, 2.2, and 2.3, we constructed a CPM dataset containing detailed information on prescriptions, indications, dosages, and other relevant data. Next, the RAG technique was used to retrieve relevant CPM information from the dataset based on the user's prompt. Subsequently, leveraging the generative capabilities of an LLM, personalized CPM recommendations were generated based on the retrieved results, ensuring that the recommendations were both accurate and highly targeted. We selected the largest parameter version of the optimal model in the TCM exam, Qwen2.5-72B-Instruct (as of October 2024), and accessed it via API using Python 3.12, setting the temperature to 0.01.

2.5. Evaluation on RAG-CPMF

2.5.1. Guidelines evaluation

In order to evaluate the accuracy and clinical appropriateness of RAG-CPMF's recommendations, we used the *clinical guideline on CPMs for dominant diseases* issued by the National Administration of TCM as the reference standard. The rationale for selecting this guideline is as follows: this guideline is oriented toward Western medical physicians and takes into consideration both Western medical diagnosis and treatment, as well as syndrome differentiation in TCM, all of which align with the core objective of RAG-CPMF—assisting Western medical physicians in appropriately prescribing CPMs. What's more, it is with higher reliability, following internationally recognized standards for evidence-based guidelines [2,7,8].

The standard data was established as follows. A total of 34 copies of the *clinical guideline on CPMs for dominant diseases*, covering 34 types of diseases, were collected by October 2024 (Supplementary Material 2). Subsequently, we invited two experts in the field of integrated traditional Chinese and Western medicine to extract three sets of data from each guideline, with each set consisting of: diseases, symptoms and signs, and the recommended CPMs. To align with clinical scenarios in which Western medicine physicians use the RAG-CPMF, the experts converted the symptoms, signs, and diseases into modern medical terms and, where appropriate, natural language, drawing on both the TCM clinical terminology system and their clinical expertise. This process resulted in a standardized dataset comprising 102 entries. Each entry links a disease with its associated symptoms and signs, expressed in modern medical terms or natural language, along with the corresponding CPM recommendations.

After constructing the standardized dataset, we compared RAG-CPMF with the baseline LLM Qwen and four widely used LLMs (GLM, ERNIE, Llama, and ChatGPT), as detailed in Table 2. The temperature parameter for all LLMs was set to 0.01. Subsequently, each entry in the standardized dataset, which consisted of a disease along with its associated symptoms and signs, was input into the LLMs, which recommended the top three most suitable CPMs. Finally, we evaluated the recommendation results. If any of the three recommended CPMs included drugs mentioned in the guidelines, the recommendation would be considered correct. Differences among models were compared using Cochran's Q test with Bonferroni correction, and the corresponding *P* values were calculated using SPSS version 27.0.

2.5.2. Clinician evaluation

Although guidelines evaluation is effective, clinician assessments offer deeper insights into the details, context, and factual accuracy within the medical domain, thereby complementing the limitations of guidelines evaluation. For the intelligent CPM recommendations, we

employed the SUS (Safety, Usability, and Smoothness) method for manual evaluation, which encompasses three dimensions: safety, usability, and smoothness [41]. The "safety" dimension assesses whether the model-generated content could mislead users or pose health risks. The "usability" dimension evaluates how effectively the content reflects professional knowledge, while the "smoothness" dimension gauges the model's overall proficiency. SUS employs a three-point scoring mechanism: a score of one denotes unacceptable performance, two signifies acceptable performance, and three represents good performance. To evaluate LLMs' performance, we enlisted three clinicians with a background in TCM to evaluate the CPM recommendations generated by the LLMs for 102 standardized cases. The Friedman test with Bonferroni correction was used to compare differences among models, utilizing SPSS version 27.0.

So as to evaluate the practical feasibility of RAG-CPMF and identify directions for future optimization, in May 2025, we conducted semi-structured interviews with clinical physicians. Following principles of information saturation, ten participants were recruited, including the three TCM clinicians from the SUS method evaluation, two additional TCM clinicians, two integrated traditional Chinese and Western medicine physicians, and three Western medicine physicians. Written informed consent was obtained from all participants, and anonymized codes were used to ensure confidentiality. The interview outline was as follows: (1) In what clinical scenarios would you consider using CPMs? (2) Do you have any suggestions for designing and applying an intelligent CPM recommendation framework in clinical practice? (3) In your clinical practice, how do you usually obtain information about CPMs? Transcripts were completed within 24 h and independently analyzed by two investigators using thematic analysis.

3. Results

3.1. CPMs data sources and processing results

The *National Drug Code Information* dataset contains 158,880 drug records. After removing duplicates, selecting CPMs, and excluding active pharmaceutical ingredients, a total of 8693 unique CPM records were retained. Among these, 8644 CPM instructions were found (Fig. 4). For instructions illustrated as image format, Optical Character Recognition (OCR) [42] was applied to extract the text, followed by manual proofreading to ensure the accuracy of the results.

3.2. CPM dataset construction: results and validation

3.2.1. Accuracy evaluation of different prompts

We randomly selected 500 samples from all the 8644 instruction texts based on sample size estimation. The sample size estimation procedure is detailed in the Supplementary Material 3. The accuracy of three types of prompts in the structured information extraction task across five LLMs is shown in Fig. 5. The results indicated that the different prompt designs had an impact on the accuracy of the models in extracting structured information. Overall, FoT showed better accuracy than CoT and ToT in all models. Therefore, we selected FoT as the optimal prompt.

3.2.2. Structured information extraction results

We have randomly sampled 500 samples from the remaining 8193 texts after excluding the 500 samples extracted in the previous trial. The Multi-LLMs Validation method based on FoT prompting was then employed to extract structured information. The accuracy of consistent results from different numbers of LLMs can be seen in Fig. 6. It showed that when the extraction results from at least four LLMs fully aligned, the accuracy of structured information extraction using the Multi-LLMs Validation method reached 100 %. In addition, we observed that the extraction results from the five LLMs were completely inconsistent across eight instruction texts. Further analysis revealed that all eight

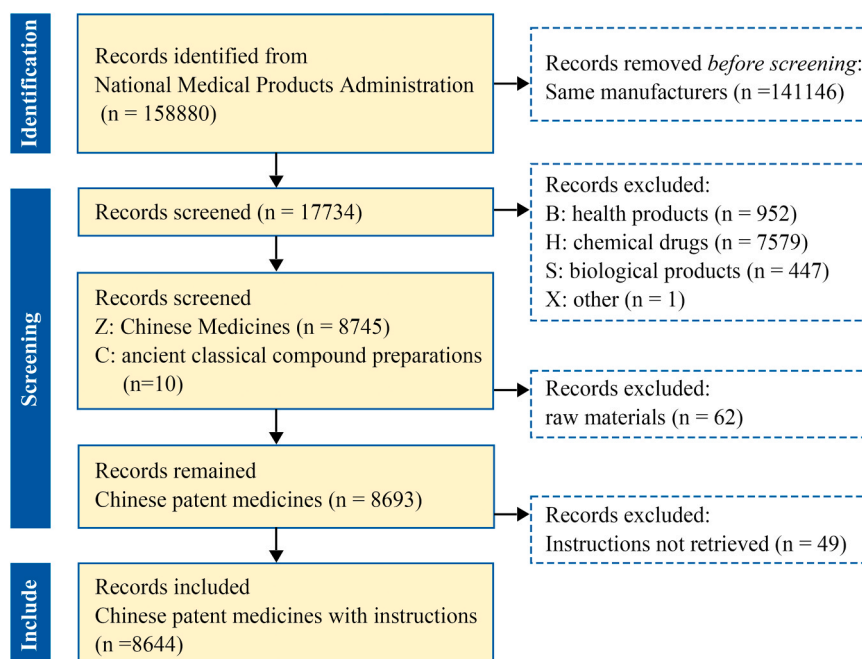


Fig. 4. This flow diagram shows the searching and selecting of CPMs. In the dataset from National Medical Products Administration, Chinese Medicines are labeled as Z, ancient classical compound preparations as C, chemical drugs as H, biological products as S, and health products as B.

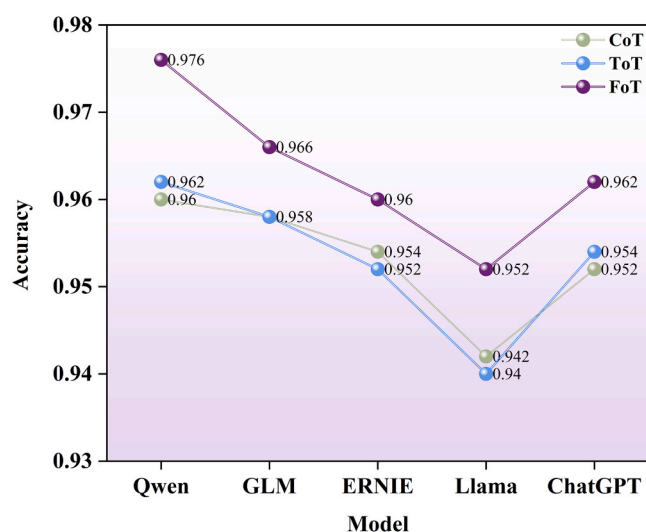


Fig. 5. Accuracy for three prompts in structured information extraction across five LLMs.

texts were instructions for ethnic patent medicines. The “Indications or functions of ethnic patent medicine” field in these instructions was written in ethnic languages, presenting challenges for LLMs in structured information extraction. The languages of different ethnic groups embody a wealth of historical, cultural, and medical knowledge, providing a solid foundation for advancing research in medical diversity. In order to ensure fidelity to the original information, the relevant fields were transcribed verbatim into the dataset, enabling researchers in ethnic medicine to further analyze and apply the data.

Based on the results of sampling test, we proposed a hypothesis, that is, provided at least four LLMs can offer consistent outcome which means the accuracy reaches 100 %, we will have no need of manual proofreading. Subsequently, we adopted this method to extract structured information covering all 8644 texts. According to this hypothesis, it was concluded that, among all 8644 texts, the number of texts in

which at least four LLMs provided consistent results was 8242, from which the accuracy of the extracted structured information was also 100 % (Fig. 6).

3.2.3. Dataset accuracy verification

After the sample size estimation, we randomly selected 500 samples from a total of 8242 texts with the accuracy of extracted structured information up to 100 %. The results extracted using the Multi-LLMs Validation method were compared with those obtained through manual extraction, and the comparison revealed an accuracy of 100 %. In other words, when at least four LLMs provide consistent results, the Multi-LLMs Validation method can indeed achieve 100 % accuracy in extracting structured information. Therefore, we just carried out manual proofreading towards the remaining 402 structured information texts, so as to greatly cut down manual workload (Fig. 6).

3.2.4. Comparison results of Multi-LLMs Validation with single LLM

We randomly selected 500 samples from a total of 8644 texts. The comparison in accuracy between Multi-LLM Validation method and traditional extraction methods of structured information based on single LLM is reflected in Fig. 7. It is obvious that the Multi-LLM Validation method outperforms the traditional extraction approach with the single model.

3.3. Workflow of intelligent CPM recommendations

Initially, Western medicine physicians collect clinical information on the patient’s disease, symptoms, and signs. Next, the patient information, along with the prompt “Based on the patient’s disease, symptoms, and signs, recommend three appropriate Chinese patent medicines. The recommendations should be well-matched to the presented clinical features.” is input into the RAG-CPMF. While the number of recommended CPMs can be adjusted as needed, three are used here as an example. To process the input, the framework utilizes the TCM clinical terminology system to standardize the input information into TCM terms based on semantic similarity calculations. The standardized terms are vectorized and matched with a pre-constructed CPM vector database using cosine similarity, and the top three most relevant CPM instructions

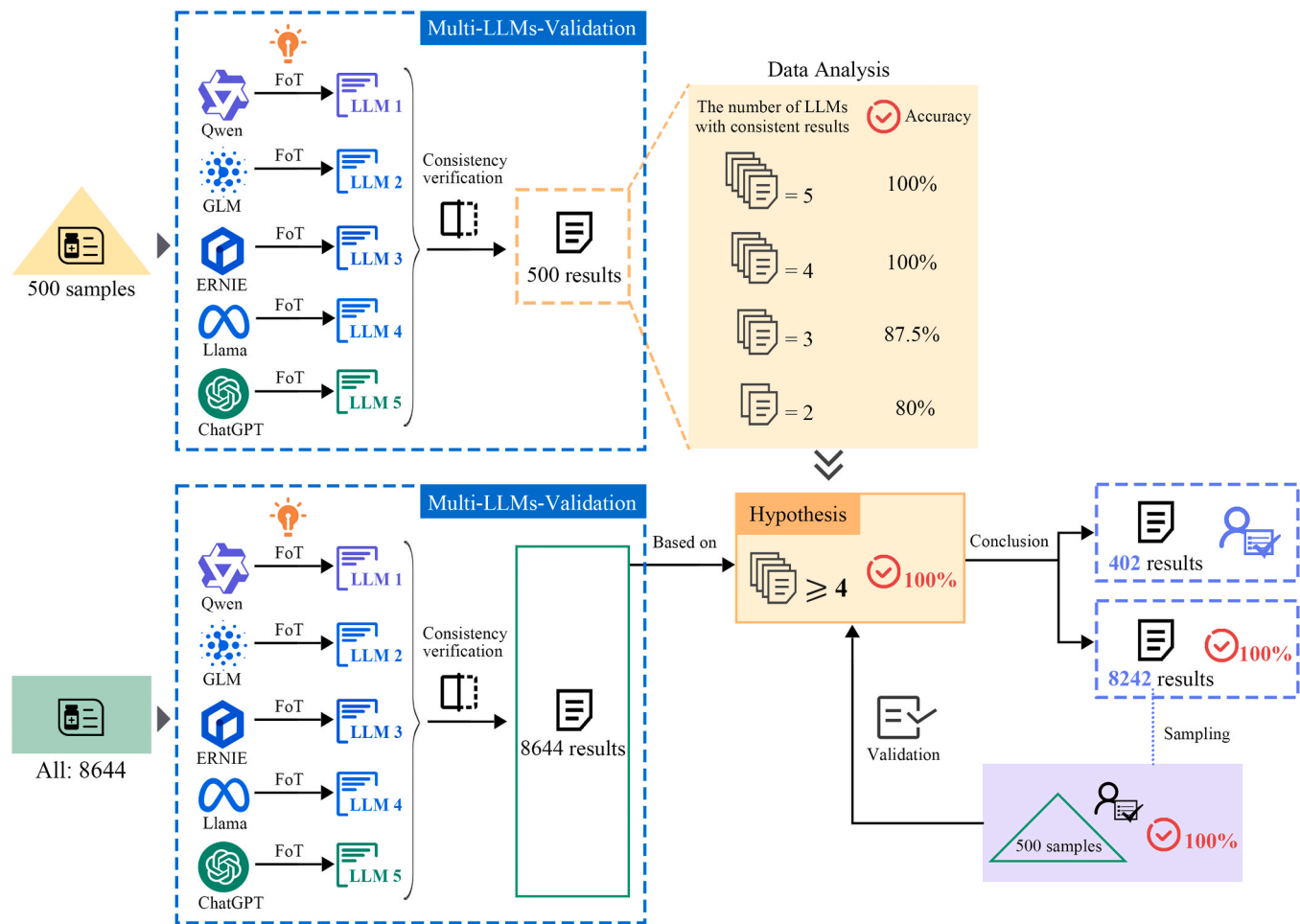


Fig. 6. Detailed results of LLMs-based structured information extraction from text. The sampling tests demonstrated that when at least four LLMs produce consistent outputs, extraction accuracy reaches 100 %. Specifically, structured information was accurately extracted from 8242 out of 8644 texts, achieving a 100 % accuracy rate. Additionally, random sampling and manual proofreading of these 8242 texts further confirmed the 100 % accuracy.

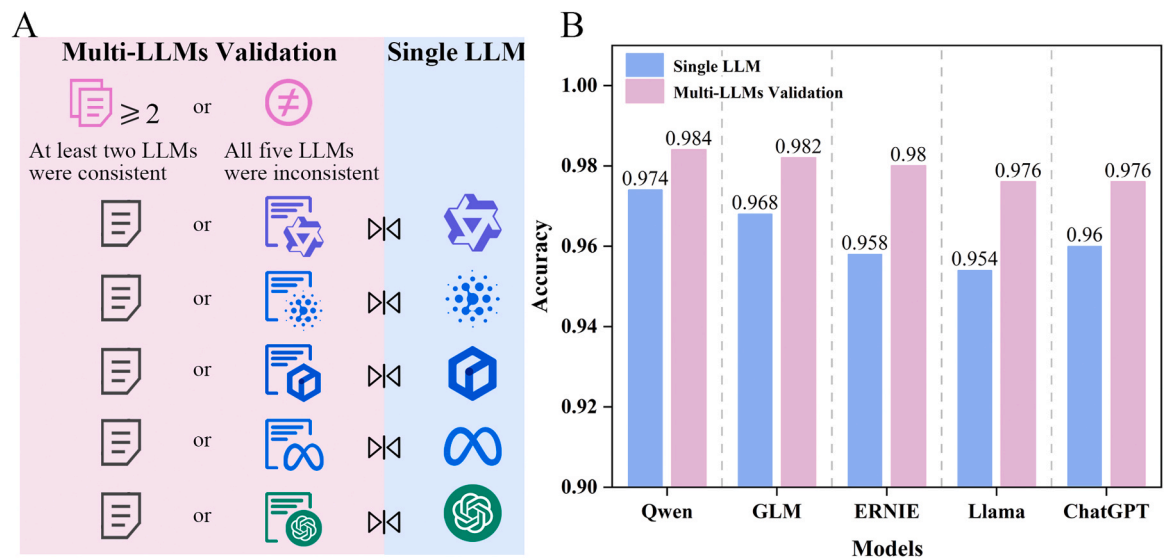


Fig. 7. Comparative analysis of the Multi-LLMs Validation method and traditional single LLM structured information extraction. (A) Methods comparison figure. (B) Comparative results figure.

are retrieved. After retrieval, the retrieved CPM instruction texts, standardized TCM terms, original clinical information of the patient, and the prompt are jointly fed into the Qwen to generate the final recommendation output. The recommended CPMs are provided with corresponding rationale and supplemented with relevant content from each CPM instruction, including indications, dosage, and other relevant information, to support clinicians in making treatment decisions (Fig. 8).

3.4. Evaluation results of RAG-CPMF

3.4.1. Quantitative evaluation results

The guidelines evaluation results are presented in Fig. 9A, illustrating the accuracy of RAG-CPMF, the baseline LLM Qwen, and four widely used LLMs (GLM, ERNIE, Llama, and ChatGPT) in recommending CPMs. RAG-CPMF achieved the highest recommendation accuracy (92.2 %), outperforming the baseline model Qwen (80.4 %). Furthermore, RAG-CPMF demonstrated significantly higher accuracy compared to GLM (72.5 %, $P < 0.01$), ERNIE (67.6 %, $P < 0.001$), ChatGPT (52 %, $P < 0.001$), and Llama (35.3 %, $P < 0.001$), with differences reaching statistical significance (Fig. 9B, Cochran's Q test with Bonferroni correction).

Evaluation based on guidelines showed that RAG-CPMF outperformed other LLMs in terms of objective recommendation accuracy. Subsequently, we conducted a further analysis of the eight recommendations that did not align with the guidelines and identified the following three issues. Firstly, some CPM instructions described indications and functions primarily using TCM terminology, without clearly specifying the corresponding modern medical diseases. For example, the indications for Shenling Baizhu Powder did not include any modern disease names, making it difficult for RAG-CPMF to achieve accurate matching due to the lack of specific disease information and explicit semantic cues. In addition, CPMs are often used in clinical practice to help treat complications of certain diseases. However, their instructions typically listed only the main diseases and did not mention clinical applications for complications. For example, although Shengxue Ning Tablets were recommended in guidelines for renal anemia caused by chronic kidney disease, their instructions only indicated iron

deficiency anemia and did not include information about such complications. Finally, a key issue was that some instructions failed to reflect the latest medical concepts. For example, cancer-related fatigue, a disease term introduced in 2018, was not mentioned in the instructions of relevant CPMs, indicating a delay in content updates.

In the clinician evaluation of model performance, RAG-CPMF exhibited advantages in terms of safety, usability, and smoothness metrics. Specifically, in the safety assessment, RAG-CPMF scored higher than the baseline model Qwen ($P < 0.05$) and significantly outperformed GLM, ERNIE, Llama, and ChatGPT ($P < 0.001$), with all differences being statistically significant (Fig. 9C, Fig. 9F). Regarding the usability metric, RAG-CPMF demonstrated significantly higher scores compared to Qwen, GLM, ERNIE, Llama, and ChatGPT ($P < 0.001$) (Fig. 9D, Fig. 9G). For smoothness metric, although no statistically significant differences were found between RAG-CPMF and Qwen, GLM, or ERNIE, RAG-CPMF achieved the highest proportion (79.4 %) of maximum ratings (score of three), surpassing Qwen (76.4 %), GLM (74.5 %), and ERNIE (71.6 %). Furthermore, RAG-CPMF showed significantly higher smoothness scores compared to Llama ($P < 0.01$) and ChatGPT ($P < 0.001$), with these differences being statistically significant (Fig. 9E, Fig. 9H).

According to clinical evaluations, RAG-CPMF exhibited superior safety and usability over other LLMs. Its smoothness was on par with Qwen, GLM, and ERNIE, showing no statistically significant differences. The safety evaluation results indicated that RAG-CPMF did not receive any “unacceptable” ratings, while all other LLMs had such ratings. Further examination of the “unacceptable” recommendations revealed that some LLMs suggested health products or other non-CPMs, which lack clearly defined clinical indications and may mislead users or pose potential health risks. It is worth noting that, although eight recommendations did not align with the guidelines, clinicians rated them as “acceptable” or “good”, suggesting that they may, to some extent, reflect the clinical considerations in TCM.

3.4.2. Qualitative evaluation results

Based on the content of semi-structured interviews, two main categories and nine subcategories were identified through textual analysis.

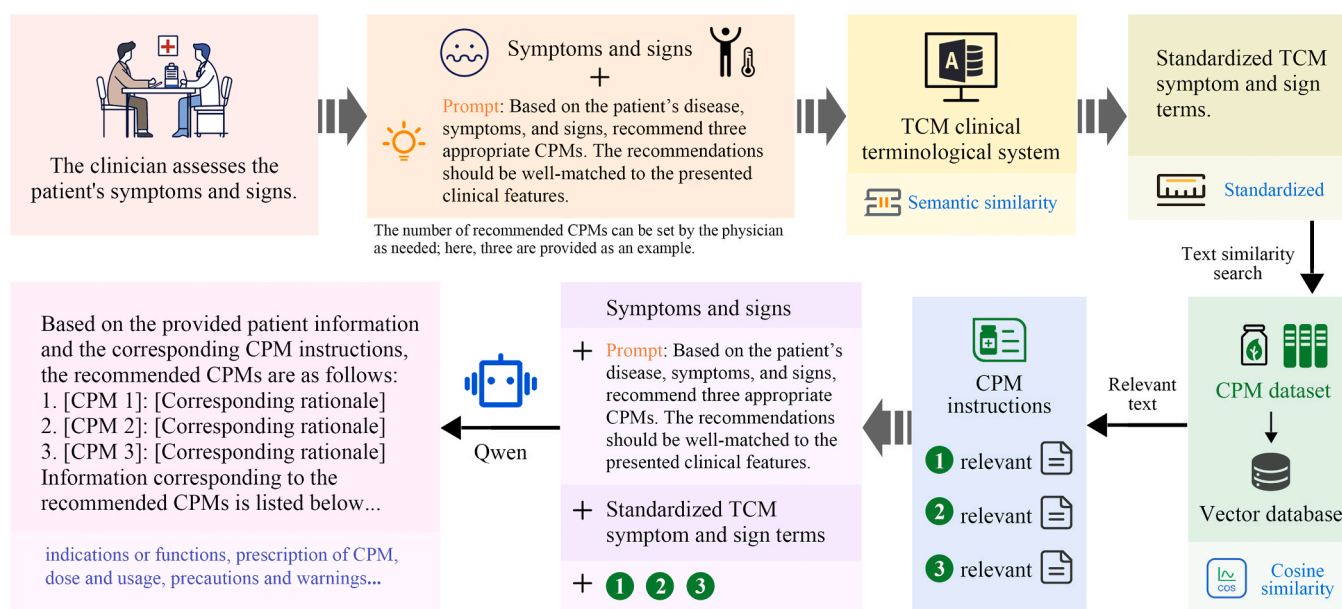


Fig. 8. Workflow of intelligent CPM recommendations. Clinical information collected by Western medicine physicians is standardized into TCM terms using a clinical terminology system. The standardized terms are vectorized and matched to a CPM vector database, retrieving the top three most relevant CPM instructions based on semantic similarity. Retrieved CPM instructions, standardized terms, original clinical data, and the prompt are input into Qwen to generate final recommendations. The output includes recommended CPMs, supporting rationale, and relevant details from CPM instructions, such as indications and dosage, to assist clinicians in treatment decision-making.

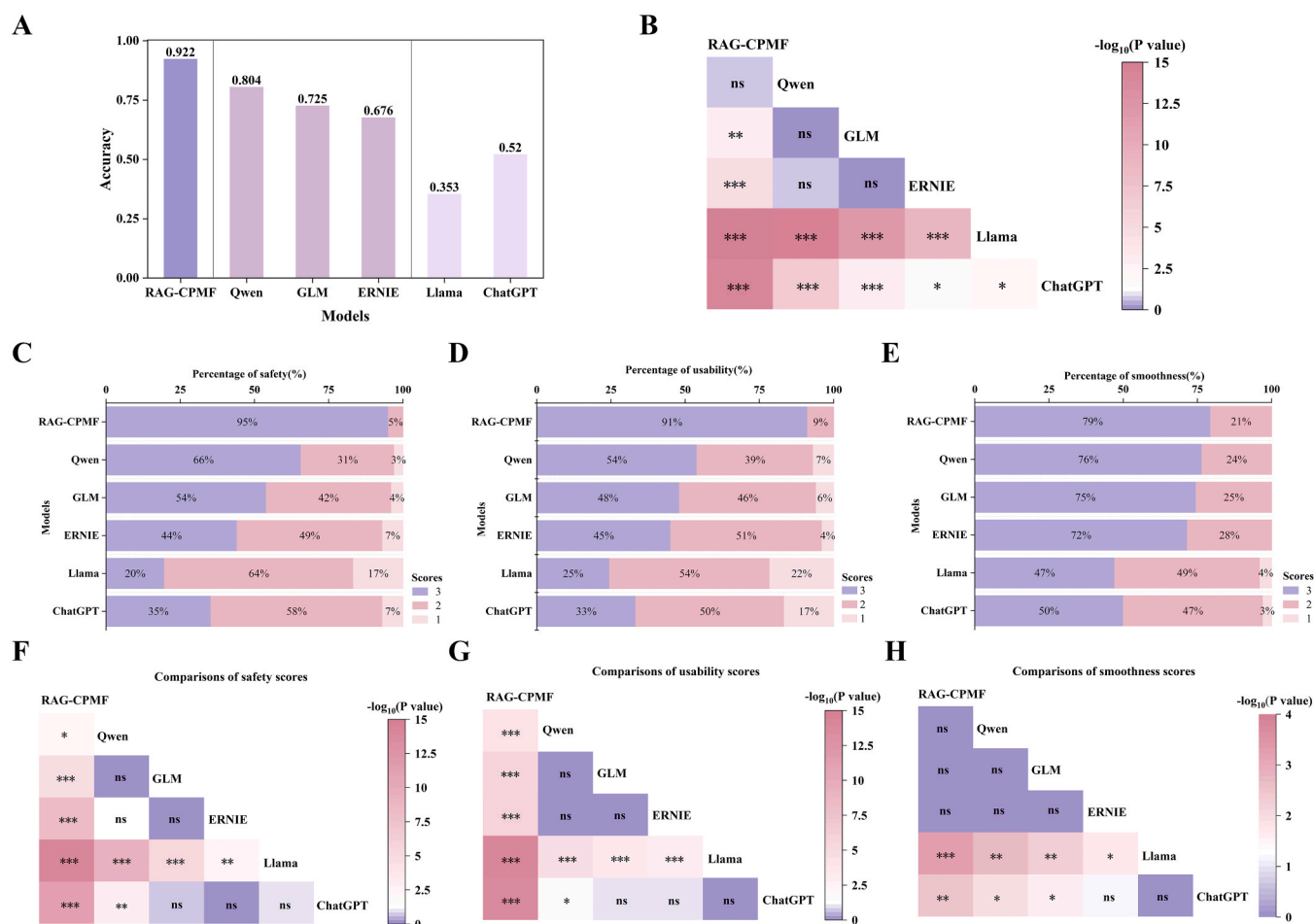


Fig. 9. Performance comparison of RAG-CPMF and other models for intelligent CPM recommendation. (A) The accuracy of guidelines evaluation. (B) Heatmap showing the P values for comparisons of accuracy between RAG-CPMF and other models. (C) The safety percentage of clinician evaluation. (D) The usability percentage of clinician evaluation. (E) The smoothness percentage of clinician evaluation. (F) Heatmap showing the P values for comparisons of safety scores between RAG-CPMF and other models. (G) Heatmap showing the P values for comparisons of usability scores between RAG-CPMF and other models. (H) Heatmap showing the P values for comparisons of smoothness scores between RAG-CPMF and other models. ns, no significance. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

For more information, refer to [Supplementary Material 4](#). Ultimately, these findings formed the core category: “Strengths and weaknesses of the intelligent CPM recommendation framework.” The strength category included five subcategories: medication safety, data quality, personalized recommendation, effectiveness of CPMs, and intelligent interaction. Its safety lay in recommending based on CPM instructions, providing information on dosage, side effects, contraindications, and other key prescribing information, while excluding non-CPM recommendations. In terms of data quality, the framework was supported by a high-quality CPM dataset characterized by comprehensive content. In terms of personalized recommendations, the framework combined Western medical diagnoses and TCM symptom patterns to deliver more tailored medication recommendations. In terms of effectiveness, CPMs demonstrated advantages in treating certain diseases and may help reduce the adverse effects of chemical drugs. In terms of intelligence, the system improved usability through natural language interaction and alleviated the operational burden on physicians.

Experts pointed out that RAG-CPMF still exhibited certain weaknesses in the completeness of data acquisition, adaptability to CPM instructions, safety of combination therapy, and support from evidence-based medicine. For more information, refer to [Supplementary Material 4](#). To begin with, RAG-CPMF still exhibited weaknesses in tongue and pulse data collection, identification of special populations, and recognition of patients with underlying diseases. Its data acquisition mechanism required further improvement. In addition, some CPM

instructions had not been updated for a long time, limiting their adaptability to current clinical needs. Furthermore, the identification of forbidden concomitant use between CPMs and chemical drugs remained limited, and systematic alerts concerning CPM incompatibilities and the cumulative risk of toxic constituents were lacking. Lastly, the evidence-based support for recommended CPMs was relatively weak, and their reliability in clinical decision-making still needed to be improved.

4. Discussion

4.1. Principal findings

When applying LLMs to CPM recommendations, the main challenge is to ensure the accuracy of the outcomes. To address this, we first proposed a Multi-LLMs Validation method to construct the largest CPM dataset, serving as an external knowledge base. Second, we introduced the RAG architecture in this context for the first time, effectively integrating the LLMs with precise retrieval from the CPM dataset, thereby developing the intelligent CPM recommendation framework, RAG-CPMF. Our main findings are as follows:

Although the existing CPM database CPMCP is commendable, it contains only 1469 CPMs and lacks a comprehensive dataset that covers all CPMs and instruction information [43]. To address this gap, we first obtained the latest and most comprehensive domestic drug information from the National Drug Code Information dataset provided by the

NMPA, ensuring that all domestic drugs are included. Next, we performed strict screening of the data, removing non-CPM items, and detailed the screening process in the results, thus obtaining the most complete and accurate CPM directory in the country. Based on this directory, we collected the latest instructions and constructed the largest CPM dataset. This dataset contains 8644 unique CPMs, each with 16 fields of instruction information. Furthermore, the dataset will be continuously updated in accordance with changes to the National Drug Code Information dataset.

In this study, we proposed the FoT prompting strategy, which aimed to enhance the accuracy of structured information extraction by decomposing the overall task into multiple mutually independent subtasks. In the field of computer science, the divide and conquer strategy improves model accuracy and robustness in complex reasoning and computer vision tasks by breaking down complex tasks into subtasks and solving them recursively [44,45]. By introducing this strategy into prompt engineering, FoT prompting divides the information extraction task into multiple subtasks, each with an independent starting point for reasoning. Each subtask is solved separately, and all results are finally aggregated to form the final output. The advantage of FoT prompting is that each subtask is responsible for only one field, which not only improves the extraction quality for individual fields, but also ensures that errors in one field do not affect others, thereby confining errors to the specific field and preventing them from impacting the overall output. However, FoT prompting introduces additional computational overhead. Therefore, in practical applications, it is necessary to balance overall performance and computational cost, and to carefully consider the granularity and number of subtasks in order to achieve optimal results. In addition, challenges remain in scenarios where texts are highly interrelated, or tasks exhibit sequential dependencies. These issues cannot be addressed by simple task decomposition alone and require further research.

The Multi-LLMs Validation method proposed in this study aims to enhance the accuracy and stability of structured information extraction through cross-model consistency validation. In systematic reviews, it was shown that independent extraction and cross-checking by multiple researchers can improve data accuracy and reduce bias [28]. Clinical studies have shown that combining multiple diagnostic techniques through back-to-back application can effectively reduce the omissions that may arise from a single method, thereby improving diagnostic accuracy [46,47]. Introducing this core concept into the field of LLMs, significant differences existed among LLMs in terms of training data types, domain expertise, and model parameters, leading to complementary performance across various tasks [27]. For example, Qwen outperformed ChatGPT in tasks related to TCM, while ChatGPT performed better in tasks related to Western medicine [48]. Based on this, this study treated different LLMs as independent entities for structured information extraction and applied consistency validation to effectively reduce biases in individual models, thereby improving the accuracy and robustness of the results. However, it should be noted that when the outputs of the LLMs are highly divergent or no clear majority is reached, the accuracy of automated decisions cannot be guaranteed, and manual proofreading remains necessary. Overall, this method can improve the accuracy of structured information extraction and reduce the workload of manual proofreading to some extent, but it still cannot fully replace manual validation.

When applying LLMs to the medical field, many studies have incorporated the RAG architecture to enhance the accuracy of responses [49,50]. In our study, compared to traditional LLMs, RAG-CPMF demonstrated significant advantages in CPM recommendations. The RAG architecture retrieves relevant fragments based on vector search and text similarity, effectively capturing semantic similarities, thereby providing more accurate responses. Furthermore, when retrieving information from the CPM dataset, the RAG architecture clearly labels the information entries used, improving the interpretability of the model [24]. Additionally, the RAG architecture offers the flexibility to extend the

dataset, enabling continuous updates and expansions of the CPM dataset to keep pace with the latest developments in the medical field. This feature adapts to the ongoing evolution of medical knowledge, ensuring timely and accurate recommendations for emerging diseases.

4.2. Comparison to prior work

Multiple studies have investigated the intelligent recommendation of CPMs, mainly through three technical approaches: deep learning [51], knowledge graphs [52], and rule-based reasoning [53]. These studies have explored various implementation strategies and have laid a solid foundation for our research. While deep learning methods demonstrate strong performance with structured inputs, they remain challenged by natural language expressions and the understanding of complex semantics [51]. Knowledge graph-based approaches offer advantages in relation extraction and reasoning, but their construction often relies heavily on manual labor and face limitations in terms of data scale [52]. Rule-based reasoning methods exhibit clear logic and are effective in standardized workflows; however, their generalization capacity remains insufficient in the face of complex clinical scenarios [53].

This study proposed RAG-CPMF, which demonstrated innovations in input adaptability, data structuring, and recommendation strategy. Specifically, it featured strong input adaptability, overcame the limitations of traditional form-based input and supported diverse textual expressions, such as natural language, modern medical terms, and TCM synonyms, thereby improving usability. In terms of data, a structured dataset covering 8644 CPM instructions was constructed, ensuring comprehensive support for recommendation. Its format also facilitated future incremental updates. Regarding the recommendation mechanism, a RAG approach was used, integrating semantic retrieval and contextual understanding. This approach avoided reliance on pre-defined rules or knowledge graphs, enhancing flexibility and applicability in clinical scenarios. In addition, the framework incorporated instruction content as contextual input into the LLM, which not only improved the accuracy of recommendations but also enhanced their interpretability.

4.3. Limitation

Given its foundation in evidence-based medicine and recognized authority, the *Clinical Guideline on CPMs for Dominant Diseases* was adopted as the evaluation standard for RAG-CPMF's recommendation results. However, as of October 2024, the guideline covered only 34 diseases and primarily focused on single-disease scenarios, typically recommending only one CPM per condition. As such, it did not sufficiently capture complex clinical scenarios, such as comorbidities or variations in disease stages. This limited scope may have introduced selection bias into the evaluation process, potentially affecting the generalizability of the results. In addition, although RAG-CPMF demonstrated high accuracy, it lacked real-time interaction and validation with actual clinicians and patients. A further limitation to the generalizability of the study was that it did not gender or sex issues. Consequently, while the framework showed promising theoretical applications, it had yet to be adequately validated in complex clinical settings.

Although this study collected 8644 CPM instructions and constructed a dataset, these instructions themselves had certain limitations. For example, some instructions had not been revised for a long time, and their indications were not effectively aligned with modern medical terms. In addition, the guideline's recommendations included adjusting medication use based on the patient's specific symptoms, with some CPMs being used for indications not explicitly listed in their instructions. Such off-label use may have led to discrepancies between RAG-CPMF recommendations and clinical practices. Therefore, it is essential to incorporate real-world data and clinical feedback to further optimize the recommendation basis and enhance the clinical applicability of RAG-

CPMF.

4.4. Challenges and future research

Although the RAG-CPMF has made some progress in intelligent CPM recommendations, several challenges remain. First, the collection of individual information is still inadequate, making it difficult to identify special populations and underlying diseases. Second, there have been discrepancies between some CPM instructions and their actual clinical use. Furthermore, the RAG-CPMF has limited ability to identify potential interactions between CPMs and chemical drugs, as well as contraindications in CPM compatibility, and it lacks an effective drug safety alert mechanism. In addition, the recommendation process does not incorporate evidence-based medicine, which may affect its reliability in clinical decision-making.

Future research should focus on enhancing system functionality and optimizing the integration of data resources. To begin with, data collection methods should be improved by adding multimodal data such as tongue diagnosis, pulse diagnosis, and voice analysis to overcome the limitations of single inquiry data. Meanwhile, the integration of electronic medical records will help improve the identification accuracy of special populations and patients with underlying diseases. Moreover, a clinical data platform integrating real-world data should be established, combining patient records and treatment feedback to update recommendations in a timely manner. Real-world data can also narrow the gap between CPM instructions and clinical practice and improve the model's ability to handle complex cases. Furthermore, introducing tools such as pharmacological knowledge graphs, drug interaction models [54], and network toxicology [55] can greatly improve the identification of risks in combination therapies. Lastly, an evidence-based medicine database should be built, including guidelines, systematic reviews, meta-analyses, and randomized controlled trials, and combined with real-world data and literature mining. This will help ensure that recommended CPMs are both consistent with TCM theory and supported by evidence-based medicine.

RAG-CPMF has certain prospects for application across different traditional medical systems. However, it may still encounter various challenges in this process, such as adaptation when extending the framework to Korean medicine [56], Tibetan medicine [57], and others. First, there may be difficulties in language processing, such as language differences and terminology mapping, especially for multilingual natural language processing. Second, systems like Korean medicine and Tibetan medicine have unique theories and drug compatibility rules, which can make it hard to integrate knowledge [56,57]. To address these challenges, it is important to develop an ontology framework that supports multiple medical systems and build a multilingual medical knowledge base, so that terms and meanings from different systems can be effectively connected and translated.

The ethical and regulatory issues of RAG-CPMF are important, especially in the medical field. Key concerns include data privacy, intellectual property, system transparency, and accountability. First, it is necessary to follow data privacy regulations, ensure that patient data are anonymized and encrypted, and prevent unauthorized access. Second, the decision-making process of RAG-CPMF should be transparent so that patients understand the role of AI in medical decisions [15]. Third, it is important to prevent AI bias and ensure fairness for all groups. Developers should also clarify data usage permissions and make data processing transparent to avoid misuse. Finally, because LLMs can change over time, continuous monitoring is needed to regularly assess their accuracy and impact on patients [58].

5. Conclusion

This study innovatively proposes the Multi-LLMs Validation method, significantly reducing manual proofreading in structured data extraction. Based on this method, the largest publicly available CPM dataset

was successfully established and continuously updated online. Furthermore, for the first time, we integrated the RAG architecture with the CPM dataset, effectively combining the generative capabilities of LLMs with external data retrieval to develop an intelligent CPM recommendation framework. Evaluation against clinical guidelines confirmed that RAG-CPMF substantially enhanced recommendation accuracy compared with general LLMs, thus providing novel insights for future research in related fields.

CRediT authorship contribution statement

Suyang Qin: Conceptualization, Writing – original draft. **Yifan Wang:** Conceptualization, Methodology. **Tangming Cui:** Data curation. **Jinge Ma:** Visualization. **Xin Zhou:** Formal analysis. **Xinglin Guo:** Formal analysis. **Chuchu Zhang:** Validation. **Chongyun Zhou:** Software. **Rongjuan Guo:** Supervision, Funding acquisition. **Haiyan Li:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

This study focuses on extracting structured data using large language models, employing five different LLMs as detailed in the methods. The writing process was conducted without AI assistance, and the authors assume full responsibility for the content of the published work.

Declaration of Competing Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Acknowledgements

This work was supported by Scientific and Technological Innovation Project of China Academy of Chinese Medical Sciences (No. CI2021B002) and National Key Research and Development Program of China (No.2021YFC2501405).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.phrs.2025.107883](https://doi.org/10.1016/j.phrs.2025.107883).

Data Availability

The CPM dataset is available at <https://gitee.com/tcmdoc/cpm> The traditional Chinese medicine clinical terminological system can be accessed at <http://tcmcts.cintcm.com>.

References

- [1] M. Xiao, J. Tian, Y. Zhou, X. Xu, X. Min, Y. Lv, M. Peng, Y. Zhang, D. Yan, S. Lang, Q. Zhang, A. Fan, J. Ke, X. Li, B. Liu, M. Jiang, Q. Liu, J. Zhu, L. Yang, Z. Zhu, K. Zeng, C. Li, Y. Zheng, H. Wu, J. Lin, F. Lian, X. Li, X. Tong, Efficacy of Huoxiang Zhengqi dropping pills and Lianhua Qingwen granules in treatment of COVID-19: a randomized controlled trial, *Pharm. Res* 161 (2020) 105126.
- [2] X. Li, L. Huang, L. Wang, X. Jin, Q. Zhou, Y. Ma, Q. Wang, T. Wang, X. Hu, D. Ye, Y. Mao, Z. Xie, Y. Chen, C. Wen, The reporting checklist for Chinese patent medicine guidelines: RIGHT for CPM, *Pharm. Res* 199 (2024) 107015.
- [3] H.A. Karunajeewa, L. Manning, I. Mueller, K.F. Ilett, T.M. Davis, Rectal administration of artemisinin derivatives for the treatment of malaria, *Jama* 297 (21) (2007) 2381–2390.
- [4] Y. Tu, Artemisinin-a gift from traditional Chinese Medicine to the World (Nobel Lecture), *Angew. Chem. Int. Ed. Engl.* 55 (35) (2016) 10210–10226.
- [5] D. Cyranoski, Why Chinese medicine is heading for clinics around the world, *Nature* 561 (7724) (2018) 448–450.
- [6] Y. Yang, X. Li, G. Chen, Y. Xian, H. Zhang, Y. Wu, Y. Yang, J. Wu, C. Wang, S. He, Z. Wang, Y. Wang, Z. Wang, H. Liu, X. Wang, M. Zhang, J. Zhang, J. Li, T. An, H. Guan, L. Li, M. Shang, C. Yao, Y. Han, B. Zhang, R. Gao, E.D. Peterson,

- Traditional Chinese Medicine Compound (Tongxinluo) and clinical outcomes of patients with acute myocardial infarction: the CTS-AMI randomized clinical trial, *Jama* 330 (16) (2023) 1534–1545.
- [7] M. Wang, H. Liu, Y. Chen, J. Yu, J. Lin, Z. Sun, J. Li, G. Bai, F. Wen, Y. Xie, J. Wang, J. Yang, S. Li, Guideline on treating community-acquired pneumonia with Chinese patent medicines, *Pharm. Res.* 196 (2023) 106919.
 - [8] L. Wu, Y. Chen, Y. Ma, Z. Yang, N. Yang, W. Deng, Y. Chen, Y. Sun, Y. Li, L. Lin, Clinical practice guideline on treating influenza in adult patients with Chinese patent medicines, *Pharm. Res.* 160 (2020) 105101.
 - [9] J. Cummings, The role of neuropsychiatric symptoms in research diagnostic criteria for neurodegenerative diseases, *Am. J. Geriatr. Psychiatry* 29 (4) (2021) 375–383.
 - [10] S.H. Siddiqi, S.F. Taylor, D. Cooke, A. Pascual-Leone, M.S. George, M.D. Fox, Distinct symptom-specific treatment targets for circuit-based neuromodulation, *Am. J. Psychiatry* 177 (5) (2020) 435–446.
 - [11] X. Gan, Z. Shu, X. Wang, D. Yan, J. Li, S. Ofaim, R. Albert, X. Li, B. Liu, X. Zhou, A. L. Barabási, Network medicine framework reveals generic herb-symptom effectiveness of traditional Chinese medicine, *Sci. Adv.* 9 (43) (2023) eadh0215.
 - [12] H. Naveed, A.U. Khan, S. Qiu, M. Saqib, S. Anwar, M. Usman, N. Barnes, A.S. Mian, A Comprehensive Overview of Large Language Models, arXiv preprint arXiv: 2307.06435.
 - [13] C. Peng, X. Yang, A. Chen, K.E. Smith, N. PourNejatian, A.B. Costa, C. Martin, M. G. Flores, Y. Zhang, T. Magoc, G. Lipori, D.A. Mitchell, N.S. Ospina, M.M. Ahmed, W.R. Hogan, E.A. Shenkman, Y. Guo, J. Bian, Y. Wu, A study of generative large language model for medical research and healthcare, *NPJ Digit Med* 6 (1) (2023) 210.
 - [14] J.A. Omiye, H. Gui, S.J. Rezaei, J. Zou, R. Daneshjou, Large language models in medicine: the potentials and pitfalls: a narrative review, *Ann. Intern Med* 177 (2) (2024) 210–220.
 - [15] B. Meskó, E.J. Topol, The imperative for regulatory oversight of large language models (or generative AI) in healthcare, *NPJ Digit Med.* 6 (1) (2023) 120.
 - [16] K. Singhal, S. Azizi, T. Tu, S.S. Mahdavi, J. Wei, H.W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl, P. Payne, M. Seneviratne, P. Gamble, C. Kelly, A. Babiker, N. Schärli, A. Chowdhery, P. Mansfield, D. Demner-Fushman, Y.A. B. Aguiar, D. Webster, G.S. Corrado, Y. Matias, K. Chou, J. Gottweis, N. Tomasev, Y. Liu, A. Rajkomar, J. Barral, C. Semturs, A. Karthikesalingam, V. Natarajan, Large language models encode clinical knowledge, *Nature* 620 (7972) (2023) 172–180.
 - [17] P. Webster, Six ways large language models are changing healthcare, *Nat. Med* 29 (12) (2023) 2969–2971.
 - [18] C. Stokel-Walker, R. Van Noorden, What ChatGPT and generative AI mean for science, *Nature* 614 (7947) (2023) 214–216.
 - [19] M. Májovský, M. Černý, M. Kasal, M. Komarc, D. Netuka, Artificial intelligence can generate fraudulent but authentic-looking scientific medical articles: Pandora's box has been opened, *J. Med Internet Res* 25 (2023) e46924.
 - [20] E.A.M. van Dis, J. Bollen, W. Zuidema, R. van Rooij, C.L. Bockting, ChatGPT: five priorities for research, *Nature* 614 (7947) (2023) 224–226.
 - [21] R. Azamfirei, S.R. Kudchadkar, J. Fackler, Large language models and the perils of their hallucinations, *Crit. Care* 27 (1) (2023) 120.
 - [22] A.L. Zaleski, R. Berkowsky, K.J.T. Craig, L.S. Pescatello, Comprehensiveness, accuracy, and readability of exercise recommendations provided by an AI-based Chatbot: mixed methods study, *JMIR Med Educ.* 10 (2024) e51308.
 - [23] Y. Yoshiyasu, F. Wu, A.K. Dhand, D. Gorelik, M. Takashima, O.G. Ahmed, GPT-4 accuracy and completeness against International Consensus Statement on Allergy and Rhinology: Rhinosinusitis, *Int Forum Allergy Rhinol.* 13 (12) (2023) 2231–2234.
 - [24] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, Q. Guo, M. Wang, H. Wang, Retrieval-Augmented Generation for Large Language Models: A Survey (2023) arXiv preprint arXiv:2312.10997.
 - [25] A. Lozano, S.L. Fleming, C.C. Chiang, N. Shah, Clinfo.ai: an open-source retrieval-augmented large language model system for answering medical questions using scientific literature, *Pac. Symp. Biocomput* 29 (2024) 8–23.
 - [26] Traditional Chinese medicine — Categories of traditional Chinese medicine (TCM) clinical terminological systems, ISO Standard 19465, 2017. [Online]. Available: <https://www.iso.org/standard/64962.html>.
 - [27] L. Zhu, W. Mou, Y. Lai, J. Lin, P. Luo, Language and cultural bias in AI: comparing the performance of large language models developed in different countries on Traditional Chinese Medicine highlights the need for localized models, *J. Transl. Med.* 22 (1) (2024) 319.
 - [28] M.J. Page, D. Moher, P.M. Bossuyt, I. Boutron, T.C. Hoffmann, C.D. Mulrow, L. Shamseer, J.M. Tetzlaff, E.A. Akl, S.E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M.M. Lalu, T. Li, E.W. Loder, E. Mayo-Wilson, S. McDonald, L.A. McGuinness, L.A. Stewart, J. Thomas, A.C. Tricco, V.A. Welch, P. Whiting, J.E. McKenzie, PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews, *Bmj* 372 (2021) n160.
 - [29] J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang, B. Hui, L. Ji, M. Li, J. Lin, R. Lin, D. Liu, G. Liu, C. Lu, K. Lu, J. Ma, R. Men, X. Ren, X. Ren, C. Tan, S. Tan, J. Tu, P. Wang, S. Wang, W. Wang, S. Wu, B. Xu, J. Xu, A. Yang, H. Yang, J. Yang, J. Yang, S. Yang, Y. Yao, B. Yu, Y. Bowen, H. Yuan, Z. Yuan, J. Zhang, X. Zhang, Y. Zhang, Z. Zhang, C. Zhou, J. Zhou, X. Zhou, T. Zhu, Qwen Technical Report, 2023, arXiv preprint arXiv:2309.16609.
 - [30] T.G.A. Zeng, B. Xu, B. Wang, C. Zhang, D. Yin, D. Rojas, G. Feng, H. Zhao, H. Lai, H. Yu, H. Wang, J. Sun, J. Zhang, J. Cheng, J. Gui, J. Tang, J. Zhang, J. Li, L. Zhao, L. Wu, L. Zhong, M.-y Liu, M. Huang, P. Zhang, Q. Zheng, R. Lu, S. Duan, S. Zhang, S. Cao, S. Yang, W.L. Tam, W. Zhao, X. Liu, X. Xia, X. Zhang, X. Gu, X. Li, X. Liu, X. Liu, X. Yang, X. Song, X. Zhang, Y. An, Y. Xu, Y. Niu, Y. Yang, Y. Li, Y. Bai, Y. Dong, Z. Qi, Z. Wang, Z. Yang, Z. Du, Z.-P. Hou, Z. Wang, ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools, 2024, arXiv preprint, arXiv: 2406.12793.
 - [31] Z. Zhang, X. Han, Z. Liu, X. Jiang, M. Sun, Q. Liu, ERNIE: Enhanced Language Representation with Informative Entities, 2019, arXiv preprint arXiv:1905.07129.
 - [32] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, G. Lample, LLaMA: Open and Efficient Foundation Language Models, 2023, arXiv preprint arXiv:2302.13971.
 - [33] K. Sanderson, GPT-4 is here: what scientists think, *Nature* 615 (7954) (2023) 773.
 - [34] M.P. Polak, D. Morgan, Extracting accurate materials data from research papers with conversational language models and prompt engineering, *Nat. Commun.* 15 (1) (2024) 1569.
 - [35] L. Wang, X. Chen, X. Deng, H. Wen, M. You, W. Liu, Q. Li, J. Li, Prompt engineering in consistency and reliability with the evidence-based guideline for LLMs, *NPJ Digit Med* 7 (1) (2024) 41.
 - [36] J. Wei, X. Wang, D. Schuurmans, M. Bosma, E.H. Chi, F. Xia, Q. Le, D. Zhou, Chain of Thought Prompting Elicits Reasoning in Large Language Models, 2022, arXiv preprint, arXiv:2201.11903.
 - [37] S. Yao, D. Yu, J. Zhao, I. Shafra, T.L. Griffiths, Y. Cao, K. Narasimhan, Tree of Thoughts: Deliberate Problem Solving with Large Language Models, 2023, arXiv preprint arXiv:2305.10601.
 - [38] D. Zhou, N. Schärli, L. Hou, J. Wei, N. Scales, X. Wang, D. Schuurmans, C. Cui, O. Bousquet, Q.V. Le, E.H. Chi, Least-to-most prompting enables complex reasoning in large language models, *Elev. Int. Conf. Learn. Represent.* (2023).
 - [39] T.B. Chandra, K. Verma, B.K. Singh, D. Jain, S.S. Netam, Coronavirus disease (COVID-19) detection in Chest X-Ray images using majority voting based classifier ensemble, *Expert Syst. Appl.* 165 (2021) 113909.
 - [40] A. Yalamanchili, B. Sengupta, J. Song, S. Lim, T.O. Thomas, B.B. Mittal, M. E. Abazeed, P.T. Teo, Quality of large language model responses to radiation oncology patient care questions, *JAMA Netw. Open* 7 (4) (2024) e244630.
 - [41] C. Liu, K. Sun, Q. Zhou, Y. Duan, J. Shu, H. Kan, Z. Gu, J. Hu, CPMLetGLM: parameter-efficient fine-tuning ChatGLM with Chinese patent medicine instructions, *Sci. Rep.* 14 (1) (2024) 6403.
 - [42] N.S. Laique, U. Hayat, S. Sarvepalli, B. Vaughn, M. Ibrahim, J. McMichael, K. N. Qaiser, C. Burke, A. Bhatt, C. Rhodes, M.K. Rizk, Application of optical character recognition with natural language processing for large-scale quality metric data extraction in colonoscopy reports, *Gastrointest. Endosc.* 93 (3) (2021) 750–757.
 - [43] C. Sun, J. Huang, R. Tang, M. Li, H. Yuan, Y. Wang, J.M. Wei, J. Liu, CPMCP: a database of Chinese patent medicine and compound prescription, *Database* 2022 (2022).
 - [44] C. Lang, G. Cheng, B. Tu, J. Han, Few-shot segmentation via divide-and-conquer proxies, *Int. J. Comput. Vis.* 132 (2023) 261–283.
 - [45] S. Lee, G. Kim, Recursion of Thought: A Divide-and-Conquer Approach to Multi-Context Reasoning with Language Models. Toronto, Canada, Association for Computational Linguistics, 2023, pp. 623–658.
 - [46] K.E. van Keulen, I.S. Papanikolaou, T.W.C. Mak, P. Apostolopoulos, H. Neumann, G. Delconte, M. Furnari, Y. Peters, J.Y.W. Lau, D. Polymeros, R.W.N. Schrauwen, F. Cavalcoli, E. Koukouloti, K. Triantafyllou, J.C. Anderson, H. Pohl, D.K. Rex, P. D. Siersema, Comparison of adenoma miss rate and adenoma detection rate between conventional colonoscopy and colonoscopy with second-generation distal attachment cuff: a multicenter, randomized, back-to-back trial, *Gastrointest. Endosc.* 99 (5) (2024) 798–808, e3.
 - [47] C. Cellier, G. Perrod, C. Colas, M. Dhooge, J.C. Saurin, T. Lecomte, E. Coron, G. Rahmi, C. Savale, S. Chaussade, J. Bellanger, X. Dray, N. Bence, M. Le Rhun, J. P. Barbieux, H. Pereira, G. Chatellier, E. Samaha, Back-to-back comparison of colonoscopy with virtual chromoendoscopy using a third-generation narrow-band imaging system to chromoendoscopy with indigo carmine in patients with lynch syndrome, *Am. J. Gastroenterol.* 114 (10) (2019) 1665–1670.
 - [48] Y. Wu, Y. Zhang, M. Xu, C. Jinzhi, Y. Xue, Y. Zheng, Effectiveness of various general large language models in clinical consensus and case analysis in dental implantology: a comparative study, *BMC Med Inf. Decis. Mak.* 25 (1) (2025) 147.
 - [49] S. Mashatian, D.G. Armstrong, A. Ritter, J. Robbins, S. Aziz, I. Alenabi, M. Huo, T. Anand, K. Tavakolian, Building trustworthy generative artificial intelligence for diabetes care and limb preservation: a medical knowledge extraction case, *J. Diabetes Sci. Technol.* (2024) 19322968241253568.
 - [50] Q. Zhou, C. Liu, Y. Duan, K. Sun, Y. Li, H. Kan, Z. Gu, J. Shu, J. Hu, GastroBot: a Chinese gastrointestinal disease chatbot based on the retrieval-augmented generation, *Front Med (Lausanne)* 11 (2024) 1392555.
 - [51] Y. Huang, Patient-oriented Chinese patent medicine recommendation system, master's thesis, Beijing University of Posts and Telecommunications, Beijing, 2021, in Chinese.
 - [52] C. Ma, Z. Meng, X. Bian, Y. Li, H. Zhao, Knowledge reasoning and intelligent recommendation based on Chinese patent medicine knowledge graph, *J. Med. Inform.* 45 (4) (2024) 14–20. +51, in Chinese.
 - [53] W. Zhang, Q. Zhang, Y. Shi, X. Zhou, Y. Meng, Intelligent Chinese patent medicine recommendation system based on TCM syndrome differentiation, *Smart Health* 6 (31) (2020) 5–9., in Chinese.
 - [54] Q. Lv, G. Chen, H. He, Z. Yang, L. Zhao, K. Zhang, C.Y. Chen, TCMBank-the largest TCM database provides deep learning-based Chinese-Western medicine exclusion prediction, *Signal Transduct. Target Ther.* 8 (1) (2023) 127.
 - [55] X. Li, L. Lin, L. Pang, K. Pu, J. Fu, Y. Shen, W. Zhang, H. Xu, Y. Niu, Application and development trends of network toxicology in the safety assessment of traditional Chinese medicine, *J. Ethnopharmacol.* 343 (2025) 119480.

- [56] G. Park, Oxidative damage in Korean medicine, *Antioxidants* 11 (3) (2022).
- [57] N. Wang, H. Zhu, Geo-authentic Tibetan medicine: a traditional pharmacological resource for promoting human health and wellness, *Front Pharm.* 15 (2024) 1432221.
- [58] J.C.L. Ong, S.Y. Chang, W. William, A.J. Butte, N.H. Shah, L.S.T. Chew, N. Liu, F. Doshi-Velez, W. Lu, J. Savulescu, D.S.W. Ting, Ethical and regulatory challenges of large language models in medicine, *Lancet Digit Health* 6 (6) (2024) e428–e432.